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REGULAR PAPER

SGGA: Semantic-Guided Generative Augmentation for Object Detection in Highly Imbalanced Disaster Imagery

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Abstract
Object detection in disaster images is often difficult because of class imbalance and a lack of labeled samples. To solve this problem, we introduce *Semantic-Guided Generative Augmentation* (SGGA), a new method that uses semantic masks and textual prompts to generate more samples for the rare classes. SGGA creates new images by changing clean road areas into `Road-Blocked` areas using mask-based sampling and prompt-guided inpainting, making sure the new objects appear in the right places. We filter the new images using CLIP similarity and LPIPS distance, ensuring high semantic and visual quality. Experiments on the RescueNet dataset show that SGGA improves `Road-Blocked` detection by +26.2% mAP@0.5 and +29.7% recall, beating other augmentation methods. Cross-domain validation on FloodNet further demonstrates SGGA's generalizability across disaster scenarios. Furthermore, t-SNE analysis confirms strong semantic alignment between real and SGGA-generated images. SGGA offers significant advantages, including spatial precision, contextual realism, and low annotation overhead, making it particularly suitable for practical deployment in disaster scenarios and other domains where spatial priors are available.

Keywords Class imbalance · Rare object detection · Image data augmentation · Generative data augmentation · Stable diffusion · Semantic-guided inpainting · Disaster imagery

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1 Introduction

Natural disasters pose severe threats to human lives, critical infrastructure, and ecosystems, with cascading effects that can persist long after the initial impact [1–6]. In the immediate aftermath of disasters, accurate and rapid assessment of infrastructure damage—particularly road accessibility—becomes crucial for effective disaster response and recovery operations. First responders require real-time information about blocked roads to navigate efficiently to victims during the crucial *golden hours* when survival rates are highest, while humanitarian organizations depend on accessible route information to deliver essential supplies such as food, water, and medicine to affected areas. Furthermore, automated damage assessment enables rapid generation of infrastructure maps, allowing governmental and non-governmental organizations to allocate resources effectively and prioritize recovery efforts based on data-driven insights.

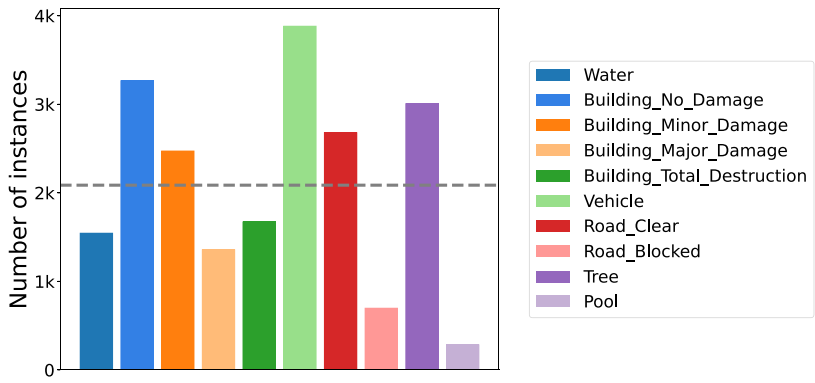

Fig. 1 The object class distribution in the RescueNet dataset [11]: Two classes (`Road-Blocked` and `Pool`) are especially rare, falling significantly below the average indicated by the gray dotted line, limiting their reliable detection for critical disaster response applications

Beyond immediate response needs, analyzing road blockage patterns from disaster imagery supports long-term urban resilience planning. City planners can identify systematically vulnerable infrastructure segments and make informed decisions about strengthening roadways, improving drainage systems, or redesigning transportation networks to enhance future disaster preparedness. To support these critical applications, recent research has increasingly employed computer vision techniques such as object detection and segmentation for disaster imagery analysis [7–9].

However, disaster images present unique challenges that significantly impact the reliability of automated detection systems. Disasters are unpredictable, rare, and geographically localized [10], leading to significant data acquisition challenges. Real disaster datasets tend to be small and severely imbalanced, with certain critical classes remaining extremely underrepresented. Figure 1 illustrates this severe class imbalance in the RescueNet dataset [11], where the `Road-Blocked` class—essential for the emergency response applications described above—contains fewer than 800 instances compared to approximately 4,000 instances of the most frequent class, `Vehicle`.

Beyond roads, other disaster-assessment tasks exhibit similar skews. In building and flood-related assessments, intact or non-flooded structures typically dominate, whereas damaged or flooded ones are scarce but operationally crucial for search-and-rescue and safety assessments. This pattern appears broadly across datasets, including *FloodNet* [12], where non-flooded categories often comprise the majority while flooded/damaged classes are underrepresented.

The challenge of detecting rare classes in imbalanced datasets is well-documented in the computer vision literature [13, 14]. Class imbalance particularly affects the performance of the minority classes, as learning algorithms tend to be biased toward majority classes during training [15, 16]. In disaster imagery, this problem is further exacerbated by the visual similarity between certain classes. The `Road-Blocked` class presents a particularly challenging case, as it shares similar global road contexts with `Road-Clear` instances while differing only in local obstacles such as debris, collapsed structures, or overturned vehicles. Figure 2 demonstrates this visual ambiguity, highlighting the subtle yet critical distinctions that emergency responders must identify to avoid impassable routes.

Traditional data augmentation techniques, from simple geometric transformations to advanced generative approaches using GANs and diffusion models, have shown promise across various domains [17–29]. However, existing generative methods face significant challenges when applied to rare and visually ambiguous classes in disaster imagery. Standard diffusion models typically require large amounts of training data to learn complex visual patterns [30], making them less effective for minority classes with limited samples. Furthermore, the lack of precise spatial control in conventional generative augmentation can result in unrealistic or contextually inconsistent synthetic images [31], particularly problematic for applications requiring high reliability such as disaster response.

While existing semantic-guided approaches such as *SPADE* [32], LaMA [33], and InstructPix2Pix [34] have demonstrated impressive capabilities in controlled generation tasks, they lack the targeted synthesis precision required for addressing severe class imbalance in disaster scenarios. These methods either rely on global semantic layouts or text-only guidance, making them insufficient for the localized, context-aware augmentation needed for rare but critical disaster-related classes.

To address these critical limitations, we propose *Semantic-Guided Generative Augmentation* (SGGA), a novel framework that introduces a dual-guidance mechanism for generating high-quality synthetic disaster imagery. Our key innovation lies in the strategic combination of semantic masks for precise spatial control and textual prompts for fine-grained content control, enabling targeted synthesis of rare-class instances with unprecedented accuracy and contextual coherence. Unlike existing methods that rely solely on text prompts or lack spatial precision, SGGA's dual-guidance approach ensures that generated images maintain both the

(a) Samples in the Road-Clear class

(b) Samples in the Road-Blocked class

Fig. 2 Samples in the `Road-Clear` and `Road-Blocked` classes in the RescueNet dataset: While both classes share similar road contexts, `Road-Blocked` scenes contain critical obstructive elements that emergency responders must identify to avoid impassable routes

correct spatial layout and semantically appropriate content for critical applications such as road accessibility and flood-related assessment in disaster scenarios.

Our main contributions are:

- *A novel dual-guidance synthesis mechanism*: We introduce the first framework that strategically combines semantic masks for precise spatial control with textual prompts for fine-grained content control, specifically designed for targeted synthesis of rare-class instances in imbalanced datasets.
- *A complete quality-controlled generative pipeline*: we develop a comprehensive framework that extends beyond image generation to include rigorous quality assessment using CLIP semantic consistency and LPIPS perceptual similarity metrics, ensuring high-fidelity synthetic data suitable for reliable downstream training.
- *Comprehensive cross-domain validation*: We provide extensive evaluation across multiple disaster domains—road infrastructure (RescueNet) and flood-related assessment (FloodNet)—demonstrating SGGA's generalizability and superior performance compared to semantic-guided baselines including Pix2Pix, with substantial improvements for rare classes (+26.2% mAP@0.5 for road blockage and consistent gains on FloodNet; see Sect. 4).

The remainder of this paper is organized as follows. Section 2 reviews previous approaches from conventional image augmentation to generative augmentation with semantic guidance. Section 3 introduces the proposed SGGA framework in detail with an example scenario of the RescueNet dataset. Section 4 presents comprehensive experiments including cross-domain validation on both road infrastructure (RescueNet) and flood-related assessment (FloodNet), comparisons with semantic-guided baselines, image quality assessment, and object detection performance across multiple disaster scenarios. Finally, Sect. 5 summarizes our work and suggests directions for future research.

2 Related Work

This section reviews previous image data augmentation methods, ranging from conventional techniques to recent generative and semantic-guided approaches. We then highlight how our proposed SGGA framework distinguishes itself through a dual-guidance mechanism combining semantic masks and text prompts for precise, context-aware rare-class synthesis.

2.1 Conventional Data Augmentation Methods

Data augmentation is a standard practice in computer vision to improve model generalization by expanding the training dataset. Traditional operations include geometric transformations (rotation, scaling), photometric changes (brightness, contrast), and kernel-based filtering (Gaussian blur) [35]. More advanced methods such as *MixUp* [17], *CutMix* [18], and Random Erasing [36] introduced sample-level blending and occlusion, while automated policy search methods such as *AutoAugment* [37] and *RandAugment* [38] learned data-dependent augmentation strategies.

Beyond data-level efforts, some studies improved model robustness for disaster recognition through architectural designs, such as BDANet [39] with its cross-directional attention mechanism and multi-scale feature fusion, and weakly supervised CNN frameworks [40] and ResNet-based transfer

Table 1 Comparison of SGGA with prior semantic- and text-guided synthesis methods

Method	Primary goal	Guidance mechanism	Key contribution
SPADE [32]	High-fidelity image synthesis	Semantic layout map	Spatially-adaptive normalization for photorealistic generation
LaMA [33]	Large-mask inpainting	Learned structural priors (FFCs)	Resolution-robust inpainting preserving global structure
InstructPix2Pix [34]	Image editing via text instructions	Text prompts (global)	Language-conditioned image-to-image editing
SGGA (ours)	Rare-class data augmentation	Semantic mask (spatial) + text prompt (content)	Precise, context-aware generative synthesis for imbalanced datasets

SGGA uniquely combines spatial and textual control for rare-class augmentation

learning approaches [41], but these model-centric advances are orthogonal to our focus on augmenting scarce training data.

However, conventional and non-generative approaches are fundamentally limited in addressing extreme class imbalance. Their random nature fails to introduce novel semantics or rare appearances, and spatial context critical to disaster understanding can be disrupted by naive transformations.

2.2 Generative Synthesis Methods

Generative models provide a new avenue for augmentation by synthesizing novel, high-quality samples. Early efforts using GANs addressed limited-data settings in domains such as medical imaging [19] and aerial imagery [20], but often suffered from mode collapse and low controllability. Diffusion models have recently shown superior stability and fidelity. Examples include *SatSynth* [26] for satellite imagery and SinDDM [28] for medical data. Large-scale diffusion frameworks like *Imagen* [24] and CDM [42] demonstrated strong class-conditional synthesis, improving downstream tasks.

Yet, these models typically operate under global prompts and lack spatial or semantic constraints. In small or highly structured datasets such as disaster scenes, this can lead to objects being generated in implausible contexts. Furthermore, fine-tuning such models is computationally expensive and prone to domain overfitting when limited real samples are available.

2.3 Semantic- and Text-Guided Synthesis Methods

To improve contextual realism, semantic-guided synthesis integrates spatial cues. *SPADE* [32] introduced spatially adaptive normalization using layout maps, while LaMA [33] achieved robust large-mask inpainting via Fourier convolutions. In parallel, text-conditioned diffusion and editing models have enabled language-driven control. *GLIDE* [43] and *InstructPix2Pix* [34] performed image synthesis or editing via natural-language instructions, whereas *ALIA* [44] and Li et al. [45] explored language-based data diversification with semantic consistency.

While these methods excel at high-quality image synthesis, they are not tailored for task-driven augmentation in rare-class, domain-specific scenarios such as disaster imagery. Text-only guidance (e.g., InstructPix2Pix) can generate plausible variations but often lacks spatial grounding for fine-grained realism. This limitation is particularly critical in safety-sensitive domains like disaster response, where context alignment strongly influences detector reliability.

Our Approach (A Targeted Framework for Augmentation): As summarized in Table 1, prior methods primarily optimize for visual realism rather than task effectiveness. Our proposed SGGA framework differs by explicitly aligning generative synthesis with downstream detection objectives. It introduces dual guidance—semantic masks provide where to generate, while text prompts define what to generate—enabling spatially precise and semantically controlled augmentation. This yields synthetic data that not only appear realistic but also meaningfully improve detector performance for rare and safety-critical classes. By bridging semantic control and generative diversity, SGGA represents a practical and generalizable advancement for imbalance-sensitive domains such as disaster response.

3 SGGA: Semantic-Guided Generative Augmentation

3.1 SGGA Pipeline Overview

Our proposed *Semantic-Guided Generative Augmentation* (SGGA) is a novel diffusion-based generative augmentation

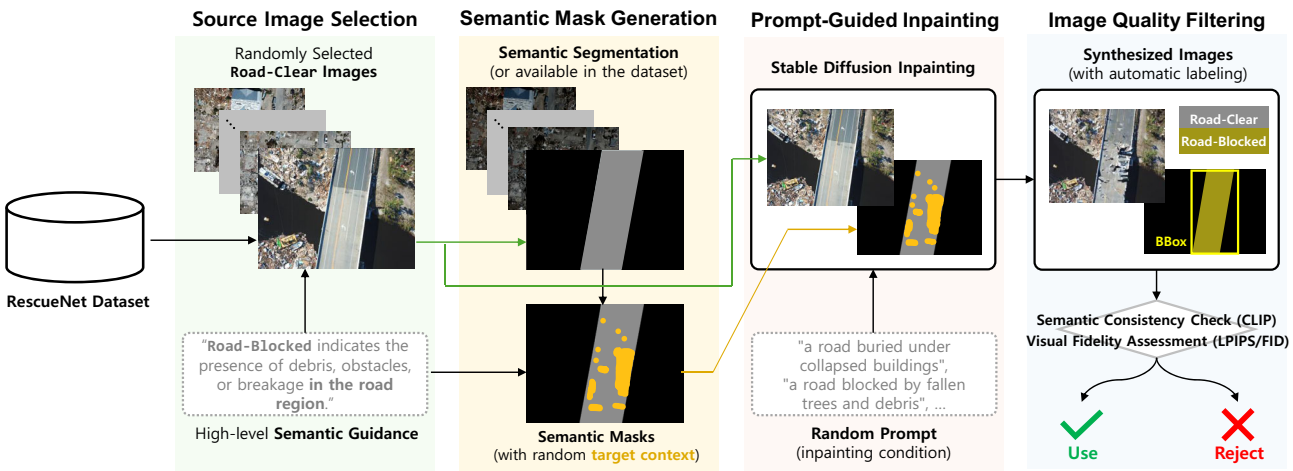

Fig. 3 Semantic-guided generative augmentation (SGGA) pipeline: at the first step, select random source images from the given training data. At the second step, generate semantic masks including random target masks. At the third step, synthesize new target images by inpainting the target masks using stable diffusion and random prompts. At the final step, semantic consistency and visual fidelity are evaluated to select high-quality images

framework that leverages semantic masks to guide spatially controlled inpainting for underrepresented class synthesis. Unlike conventional augmentation techniques or prompt-only generation approaches, SGGA can place the target context in reasonable regions supervised by the given semantic masks. SGGA synthesizes new images based on existing training images by inpainting the target context into the given image.

We will describe SGGA in more detail with respect to our target imbalanced disaster dataset, specifically for augmenting the `Road-Blocked` class in the RescueNet dataset. First, we define a high-level semantic guidance for the `Road-Blocked` class:

- `Road-Blocked` indicates the presence of debris, obstacles, or breakage *on the road region*.

Based on this semantic constraint, SGGA generates new `Road-Blocked` samples from existing `Road-Clear` samples. SGGA automatically fills the target context (e.g., debris, obstacles, or breakage) exactly on the given road region (defined in its semantic mask). The mask-conditioned inpainting ensures that the target context is located exactly on the natural and semantically reasonable region.

As illustrated in Fig. 3, the SGGA pipeline consists of four main steps:

1. *Source Image Selection:* From among the RescueNet images annotated as `Road-Clear`, we randomly sample a batch of N=200 (or some fixed number) to serve as inpainting bases.
2. *Semantic Mask Generation:* Given a source image, we obtain a road-region mask (either from the RescueNet annotations or by running a segmentation model). We then sample a random subregion within that road mask to serve as the inpainting target for `Road-Blocked` context.
3. *Prompt-Guided Inpainting:* The selected mask is fed into Stable Diffusion's inpainting module, along with one randomly chosen disaster-specific prompt (e.g., "a road buried under collapsed buildings", "a road blocked by fallen trees and debris"), to synthesize realistic obstructions within the masked region.
4. *Image Quality Filtering:* Finally, all generated candidates undergo semantic consistency checking (via CLIP), visual fidelity assessment (via LPIPS/FID), and automatic bounding-box labeling. Only samples passing these filters are retained; the rest are discarded.

3.2 Source Image Selection

Similar to conventional image data augmentation, SGGA also uses random sampling to select source images. This randomness helps maintain the underlying data distribution and prevents sampling bias, making the augmented data more generalizable.

However, SGGA does not perform completely random sampling across all training data. Instead, it uses conditional random sampling, taking into account the high-level semantic guidance for generating the target images. For example, to generate new `Road-Blocked` images, SGGA synthesizes debris, obstacles, or breakage only in clean road areas available as the `Road-Clear` class. Therefore, random sampling is applied to images that include the `Road-Clear` class.

3.3 Semantic Mask Generation

The RescueNet dataset provides pixel-wise semantic segmentation masks, which we leverage to generate binary inpainting masks for `Road-Clear` regions (class ID 7).

First, we extract the connected components corresponding to the road class and resize the resulting mask to 512×512 using nearest-neighbor interpolation. To promote diversity and avoid over-inpainting, we randomly sample approximately 60% of the road pixels to form a sparse binary mask.

This localized inpainting strategy ensures that generative edits are confined to valid spatial regions while preserving the global scene layout and semantic boundaries. As a result, our model inserts obstructions only within road areas, minimizing unintended artifacts in background or non-road objects.

3.4 Prompt-Guided Inpainting

Stable Diffusion Inpainting [46] is employed to synthesize realistic disaster-related obstructions specifically within spatially constrained mask regions—rather than across the entire image—thereby preserving global scene layout.

Each inpainting operation is guided by two components: (1) a binary mask indicating the exact region to modify (e.g., a road segment), and (2) a text prompt describing the target obstruction. As shown in the third column of Fig. 3, this combination allows SGGA to inject realistic road-blocking content within semantically relevant contexts.

It is important to clarify that this inpainting process constitutes novel visual synthesis, not merely a recombination of existing features from the training set. The power of leveraging a large, pre-trained model like Stable Diffusion lies in its vast generative prior, learned from billions of diverse images. When fine-tuned on the RescueNet dataset, the model adapts this prior to the specific context of disaster imagery, rather than simply memorizing and replicating features. Therefore, guided by a prompt and a spatial mask, the model synthesizes entirely new, plausible configurations of debris that are consistent with the disaster context but are not direct replicas of any instance in the training data. This process of creative synthesis is crucial for generating diverse and informative training samples, and we provide visual evidence for this in Appendix A.

To ensure the generated obstructions resemble realistic post-disaster scenarios, we design carefully curated prompts such as:

- *"a road buried under collapsed buildings"*
- *"a road blocked by fallen trees and debris"*
- *"a road covered with rubble from destroyed houses"*
- *"a road jammed with overturned wreckage"*
- *"a road partially obstructed by a landslide"*
- *"a cracked road surrounded by disaster debris"*
- *"a road overrun by flood debris and branches"*

To maintain spatial relevance and prevent semantic mismatches, each prompt is prefixed with *"a road"*, reinforcing road-specific context. While the segmentation mask dictates *where* content is inserted, the textual prompt controls *what* is synthesized.

This two-axis conditioning—spatial guidance via segmentation masks and semantic control via textual prompts—distinguishes SGGA from prior methods that rely solely on text-driven generation or random copy-paste augmentation. By aligning both spatial location and semantic intent, SGGA produces visually coherent and contextually appropriate obstructions, resulting in high-quality disaster scenario augmentations.

3.5 Image Quality Filtering and Analysis

To ensure the generation of high-quality synthetic data, SGGA incorporates comprehensive post-processing and validation steps combining semantic, perceptual, and latent-level analysis.

1) Semantic Consistency via CLIP

All generated images are screened using CLIP ViT-B/32 [47] to verify semantic alignment with the intended prompts. To ensure fair comparison across methods, only samples with cosine similarity scores above a fixed threshold $\tau_{\text{CLIP}} = 0.80$ are retained.

$$\mathcal{I}_{\text{final}} = \left\{ I \in \mathcal{I}_{\text{gen}} \mid S_{\text{CLIP}}(I, P) > \tau_{\text{CLIP}} \right\}, \tag{1}$$

where the semantic similarity is defined as:

$$S_{\text{CLIP}}(I, P) = \cos\left(f_{\text{img}}(I), f_{\text{text}}(P)\right) \tag{2}$$

Here, $f_{\text{img}}(I)$ and $f_{\text{text}}(P)$ represent the visual and textual embeddings obtained from the CLIP ViT-B/32 model, respectively. This ensures that only perceptually consistent images are retained for training.

2) Visual Fidelity Assessment via LPIPS and FID

We compute LPIPS [48] distances between synthetic images and real references. Samples with LPIPS values exceeding $\tau_{\text{LPIPS}} = 0.68$ are discarded.

$$\mathcal{I}_{\text{final}} = \left\{ I \in \mathcal{I}_{\text{gen}} \mid D_{\text{LPIPS}}(I, I_{\text{real}}) \leq \tau_{\text{LPIPS}} \right\} \tag{3}$$

The selection of these thresholds, $\tau_{\text{CLIP}} = 0.80$ and $\tau_{\text{LPIPS}} = 0.68$, is based on an analysis of the trade-off between sample quality and generation yield. Stricter thresholds produced higher-quality images but resulted in a very low acceptance rate, making data generation inefficient. Conversely, looser thresholds increased the yield but allowed the

inclusion of samples with visual artifacts that could potentially degrade training performance. Our chosen values thus represent a practical balance, ensuring high semantic and visual quality while maintaining a reasonable generation yield.

We also compute the Fréchet Inception Distance (FID) [49], which measures the similarity between the distributions of features extracted from an Inception-v3 network for real and generated images. Let (μ_r, Σ_r) and (μ_g, Σ_g) denote the mean and covariance matrices of Inception features for real and generated images, respectively. FID is then calculated as:

$$\text{FID} = |\mu_r - \mu_g|_2^2 + \text{Tr}(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2}) , \quad (4)$$

where $\text{Tr}(\cdot)$ denotes the trace of a matrix.

Although we report FID to quantify the distributional gap between synthetic and real samples, we do not use it as a filtering criterion. Instead, final image selection is governed by thresholds on CLIP similarity and LPIPS distance. In future work, one may impose a FID threshold (e.g., $\tau_{\text{FID}} \leq 0.70$ when scaled by 0.01) to further exclude distributionally out-of-domain samples.

3) Latent Space Validation via t-SNE

To qualitatively assess high-level semantic alignment between real and generated images, we visualize the CLIP embeddings using t-distributed Stochastic Neighbor Embedding (t-SNE) [50] (see Sect. 4.3). Let $E = \{f_{\text{CLIP}}(I) \mid I \in \mathcal{I}_{\text{real}} \cup \mathcal{I}_{\text{gen}}\}$ be the set of CLIP embeddings. We project this high-dimensional embedding space into 2D using t-SNE:

$$\hat{E} = \text{t-SNE}(E;\ \text{perplexity} = 20) . \quad (5)$$

We set the perplexity to 20 following standard heuristics, where $\sqrt{n} \leq \text{perplexity} \leq n/5$ for $n = 400\text{-}600$ total samples. With 200 images per category (Real, SGGA, and optionally LoRA), this setting provides a stable projection granularity.

4) Manual Verification and Label Reassignment

After image generation, labels are automatically reassigned from `Road-Clear` (class 7) to `Road-Blocked` (class 8) via a custom contour-based method to ensure semantic correctness. All generated bounding boxes are then manually verified using the Roboflow annotation platform to ensure label correctness and compatibility with YOLOv8 training requirements [51].

While this pipeline is demonstrated using `Road-Clear` to `Road-Blocked` conversion, the SGGA framework is broadly applicable to other underrepresented classes where spatial priors can be extracted via segmentation.

5) Impact on Class Distribution

The incorporation of 200 high-quality synthetic images improved the representation of the `Road-Blocked` class, helping to mitigate the original class imbalance and enhance model robustness for mission-critical categories [13].

4 Experiment

We conducted comprehensive experiments to evaluate the effectiveness of our proposed SGGA framework for addressing class imbalance in disaster object detection. The experiments systematically assess both the quality of generated synthetic images and their impact on object detection performance, using CLIP/LPIPS/FID for image quality and mAP@0.5/mAP@0.5:0.95 (YOLOv8n) for detection.

Our evaluation focuses on the severely underrepresented `Road-Blocked` class in the RescueNet dataset (3.3% of total instances) and includes cross-domain validation on *FloodNet* (*Non-Flooded*→*Flooded*) to test generalizability [12]. Across all studies, we compare SGGA against Conventional Augmentation, InstructPix2Pix [34], and Diffusion with LoRA [52]. These experiments demonstrate that SGGA produces plausible, context-aware rare-class samples that significantly improve detection accuracy without degrading overall performance.

4.1 Disaster Dataset

The RescueNet dataset was designed for semantic segmentation tasks in disaster response scenarios [11]. This dataset contains 4494 high-resolution UAV images, each with dimensions of 4000 × 3000 pixels. For real-time disaster response applications, object detection offers superior computational efficiency and deployment feasibility compared to pixel-wise segmentation approaches [53]. To adapt the dataset for object detection, we converted pixel-wise semantic masks into bounding box annotations using a custom OpenCV-based contour extraction algorithm, preserving the original data partitioning (80% training/10% validation / 10% testing) [11].

1) Class Imbalance Analysis

Analysis of the RescueNet dataset revealed a pronounced class imbalance. Notably, the `Road-Blocked` class accounts for only 3.3% (698 out of 20,867) of total annotated instances. This makes it a severely imbalanced but operationally critical class. Full class statistics are presented in Fig. 1 in Sect. 1. This distributional skew, exacerbated by limited spatial coverage of minority classes, aligns with documented challenges in imbalanced learning scenarios [13].

As shown in Table 4, preliminary YOLOv8n experiments revealed the impact of this imbalance: `Road-Blocked` class

achieved just 30% precision and mAP@0.5 below 0.35, confirming the necessity for targeted augmentation strategies. These findings underscore the necessity of targeted data augmentation strategies to improve detection performance for low-frequency but operationally vital classes.

2) Cross-Domain Dataset (FloodNet)

To assess cross-domain generalizability, we additionally use the *FloodNet* dataset, which provides high-resolution UAV imagery with semantic segmentation labels for post-flood scene understanding [12]. FloodNet includes 10 classes distinguishing between flooded and non-flooded structures, including *Road Flooded* and *Road Non-Flooded* categories. For object detection compatibility, we derive bounding boxes from segmentation masks and focus on the binary transformation (*Road Non-Flooded*→*Road Flooded*) that mirrors the rare-class emphasis in RescueNet. Following the same protocol, we keep validation/test splits fixed and apply augmentation only to the training split (80%/10%/10% unless otherwise specified). Consistent with prior reports, FloodNet exhibits strong class imbalance with *Road Non-Flooded* instances significantly outnumbering *Road Flooded* cases, making flooded road detection critical for emergency route planning.

4.2 Augmentation Configuration

1) Augmentation Strategies

To improve the detection of underrepresented classes, we propose a targeted data augmentation method that generates context-aware synthetic training samples. The challenge of detecting `Road-Blocked` objects stems from their dual nature: they are statistically rare yet operationally critical for effective disaster response.

We hypothesize that conventional augmentation techniques—such as mosaic or CutMix—introduce global stochastic variability that may obscure minority class features rather than enhancing them [35]. Accordingly, targeted and context-aware synthesis methods may offer more focused improvements for underrepresented categories [46].

Each experimental scenario utilized the same base `RescueNet` object detection dataset [11], consisting of 3,595 training images, 449 validation images, and 450 test images. To ensure consistent evaluation, the validation and test sets were held constant across all experiments.

The augmentation strategies were applied only to the training set, with all synthetic images targeting the `Road-Blocked` class. Specifically, the five distinct training configurations we designed are as follows:

1. *Baseline (No Augmentation)*: Original RescueNet dataset with 3,595 training images and no additional data.
2. *Conventional Augmentation*: Original dataset augmented with 200 `Road-Blocked` images using standard geometric and photometric transformations (e.g., flipping, scaling, brightness adjustment). These 200 images were manually generated by applying a subset of YOLOv8's typical augmentation techniques to existing `Road-Blocked` instances in the RescueNet training set, not automatically by the YOLOv8 pipeline itself. The resulting training set contains 3,795 images.
3. *Diffusion Model with LoRA*: Original dataset was combined with 200 `Road-Blocked` images generated via Low-Rank Adaptation (LoRA) fine-tuning of Stable Diffusion [52], totaling 3,795 training images.
4. *InstructPix2Pix (text-guided editing)*: Original dataset was combined with 200 `Road-Blocked` images generated via instruction-based image editing (InstructPix2Pix) [34], which transforms *Road-Clear* scenes into *Road-Blocked* scenes using natural-language prompts. The resulting training set contains 3,795 images.
5. *Segmentation-Guided Inpainting (SGGA)*: Original dataset augmented with 200 `Road-Blocked` images synthesized through semantic mask-guided diffusion inpainting, inspired by techniques such as LaMA [33], which performs structure-preserving image completion within masked regions. While LaMA is primarily designed for generic inpainting using learned structural priors, SGGA extends this concept by introducing domain-specific prompts and semantic masks to inject obstruction-specific visual elements into `Road-Clear` regions. This results in scene-consistent, disaster-relevant augmentations. For a detailed explanation of the SGGA framework and its pipeline, please refer to Sect. 3.

Unless otherwise noted, all generative configurations add exactly 200 class-targeted samples (a total of 3795 training images). All experiments utilized YOLOv8n [51] and were evaluated on an identical test set to ensure fair comparison across experiments.

2) Diffusion-Based LoRA Augmentation

To augment the dataset with context-aware `Road-Blocked` samples exhibiting high visual fidelity, we employed Stable Diffusion fine-tuned with Low-Rank Adaptation (LoRA) [52]. The fine-tuning process leveraged the RescueNet training dataset to capture disaster-specific visual characteristics and contextual patterns [46].

3) InstructPix2Pix Augmentation (Text-Guided Editing)

We additionally evaluate an instruction-based editing baseline using InstructPix2Pix [34] to convert *Road-Clear* images into *Road-Blocked* scenes via natural-language prompts (no paired supervision or discriminator training is required). We apply the released editing pipeline with standard settings (classifier-free guidance with text guidance scale sT

= 7.5, image guidance scale sI = 1.2) and generate 200 edited images from training-split *Road-Clear* scenes. To prevent leakage, editing is performed strictly within the training split. For FloodNet, we mirror this setup with *Road Non-Flooded→Road Flooded* edits using flood-specific prompts such as "add flood water to cover the road."

4) Prompt Generation and Curation

Prompts for LoRA fine-tuning and image generation were automatically generated from the RescueNet training images using a BLIP-2 model (Salesforce/blip2-opt-2.7b). A multistage cleaning and filtering pipeline was used to ensure semantic relevance and quality. This pipeline removed redundant tokens, semantically enhanced vague expressions (e.g., "a lot of debris" to "scattered debris"), and filtered captions based on the presence of required disaster-relevant keywords (e.g., "debris", "road", "tree") and the exclusion of irrelevant terms (e.g., "bird", "beach"). Captions shorter than 4 tokens were also discarded. For cross-domain experiments on FloodNet, we adapted prompts to flood-specific scenarios by replacing debris-related terms with flood descriptors (e.g., "debris on road" to "flood water covering road", "blocked by rubble" to "submerged under flood water") while applying the same filtering rules. InstructPix2Pix applied the same editing procedure for *Road Non-Flooded→Road Flooded* using flood-centric prompts such as "add flood water to cover the road surface."

5) Training Parameters

The Stable Diffusion U-Net was fine-tuned using the AdamW optimizer with a learning rate of 1×10^{-5}. Training was configured with a batch size of 1 and an image resolution of 512×512 pixels. An early-stopping mechanism with a patience of 3 epochs was implemented to prevent overfitting, based on a composite score:

$$Q = \mathrm{CLIP}(x, \hat{x}) - \mathrm{LPIPS}(x, \hat{x}), \tag{6}$$

where x is the original image and $\hat{x}$ is the generated image. All training seeds were set to 42 for reproducibility, and inference used randomized seeds, a guidance scale of 8.0, and 80 sampling steps. For InstructPix2Pix, we followed the released editing pipeline with classifier-free guidance and editing strength selected on the validation set (default: 50 sampling steps). All data generation and model fine-tuning were performed on an NVIDIA GeForce RTX 3090 GPU (24GB VRAM). The average time to generate a single image using this hardware setup was approximately 4.5 s.

6) Sample Generation

We generated 200 synthetic `Road-Blocked` images using a predefined set of disaster-relevant prompts that emphasize aerial perspectives and structural damage. During inference, one prompt was randomly selected per image to promote diversity across samples. Example prompts include:

- *"an aerial view of a destroyed road and buildings"*
- *"an aerial view of a destroyed road and cars"*
- *"a view of a building and a road that has been destroyed"*
- *"an aerial view of houses and a road that have been destroyed"*
- *"an aerial view of houses and a road that were destroyed by Hurricane Florence"*

For FloodNet cross-domain validation, we generated 200 synthetic `Road Flooded` images using flood-specific prompts:

- *"an aerial view of a flooded road covered with water"*
- *"an aerial view of flooded streets after heavy rainfall"*
- *"a view of roads submerged under flood water"*
- *"an aerial view of houses and roads flooded with muddy water"*
- *"an aerial view of roads and buildings affected by flood water"*

In parallel, 200 *Road-Clear→Road-Blocked* edits were produced from training-split images using InstructPix2Pix for RescueNet, and 200 *Road Non-Flooded→Road Flooded* edits for FloodNet (text-guided editing without paired supervision).

4.3 Generated Image Quality Assessment

To rigorously validate our SGGA method, we employ quality control metrics (CLIP, LPIPS, FID) and latent-space visualization (t-SNE), ensuring high-quality augmentation samples as detailed in Sect. 3. This experiment evaluates the semantic and perceptual quality of synthetic `Road-Blocked` images generated through different augmentation methods, with emphasis on diffusion-based approaches. Since visual realism and semantic fidelity are critical for effective downstream training, we assess generated images using both quantitative metrics and qualitative embedding analysis.

4.3.1 Configuration

We retained only generated samples with a CLIP similarity above 0.80 and a mean pixel intensity above 5, as described in Sect. 3.5. LPIPS (≤ 0.68) was used as a secondary measure to ensure perceptual fidelity. For fairness, the same criteria were applied to *all* methods (SGGA, Diffusion+LoRA, and InstructPix2Pix). SGGA and LoRA oversampled (300 candidates → top 200 passing the thresholds), while InstructPix2Pix performed iterative edits until 200 passing images were curated.

All retained images were manually annotated using the Roboflow annotation platform, ensuring high-quality bounding box labels for the `Road-Blocked` class. This human-in-the-loop validation maintained semantic alignment and

Table 2 Quantitative comparison of image quality for generative augmentation methods on RescueNet (↑ higher is better; ↓ lower is better)

Generative method	CLIP (↑)	LPIPS (↓)	FID (↓)
InstructPix2Pix [34]	0.831	0.698	1.367
Diffusion with LoRA [52]	0.865	0.666	1.284
SGGA (ours)	0.957	0.641	1.043
Relative improvement vs. best baseline (LoRA)			
SGGA gain	+10.6%	−3.8%	−18.8%

FID values are scaled by 0.01 for readability

Table 3 Cross-domain quality evaluation on FloodNet dataset

Generative method	CLIP (↑)	LPIPS (↓)	FID (↓)
InstructPix2Pix [34]	0.824	0.672	1.289
Diffusion with LoRA [52]	0.851	0.658	1.196
SGGA (ours)	0.934	0.624	0.987
Relative improvement vs. best baseline (LoRA)			
SGGA gain	+9.8%	−5.2%	−17.5%

SGGA maintains superior performance across disaster domains (↑ higher is better; ↓ lower is better)

annotation fidelity—both essential for effective supervised learning [54].

4.3.2 Results and Discussion

1) Image Quality Evaluation Results

As demonstrated in Tables 2 and 3, SGGA consistently outperforms other methods across CLIP, LPIPS, and FID on both datasets, emphasizing superior semantic and visual quality across disaster domains. We evaluate three complementary metrics: CLIP similarity for semantic consistency, LPIPS distance for perceptual fidelity, and Fréchet Inception Distance (FID) for statistical realism. To ensure fair and statistically meaningful comparisons, all experiments were conducted under identical conditions using the same GPU (NVIDIA GeForce RTX 3090), dataset size (200 synthetic samples per method), and inference parameters. Statistical significance of metric differences (CLIP, LPIPS, FID) was verified via bootstrap resampling (n=1,000), with p-values < 0.05. Note that FID scores are scaled by a factor of 0.01 for readability.

On RescueNet, SGGA achieved substantial improvements over diffusion LoRA across all metrics: +10.6% in CLIP similarity, −3.8% in LPIPS, and −18.8% in FID, indicating better semantic consistency, perceptual fidelity, and statistical realism, respectively. Cross-domain results on FloodNet further confirm SGGA's generalizability, with consistent improvements of +9.8% CLIP, −5.2% LPIPS, and −17.5% FID over the best baseline, demonstrating robust performance across different disaster scenarios. Figure 4 corroborates these trends with clearer, context-aware obstructions across both datasets.

2) Distributional Analysis

To qualitatively assess the semantic distribution and high-level consistency between synthetic and real `Road-Blocked` images, we conducted a t-SNE visualization of CLIP embeddings [50], as shown in Fig. 5. We randomly sampled 200 images each from the real and SGGA-generated sets, and optionally from the LoRA-generated set, resulting in a total of 400–600 samples depending on configuration. For each image, a 512-dimensional embedding was extracted using CLIP ViT-B/32 and projected into 2D with t-SNE (perplexity 20). This choice of perplexity lies within the commonly recommended range for $n \in [400, 600]$ samples [50]. The visualization demonstrates that SGGA-generated embeddings form clusters tightly integrated with real instances, validating semantic alignment and effective domain adaptation.

Figure 5 illustrates that SGGA embeddings align more closely with real data and exhibit broader variation than LoRA, indicating both fidelity and diversity.

4.4 Object Detection Experiment

To rigorously evaluate the effectiveness of our proposed data augmentation strategies, particularly SGGA, we conducted comprehensive object detection experiments. This section details the experimental configuration and evaluation metrics, followed by quantitative results focusing on the underrepresented `Road-Blocked` category and overall performance.

4.4.1 Configuration

1) Model Configuration

We adopted `YOLOv8n`, a lightweight variant of the YOLO family comprising approximately 3 million parameters. It was selected for its balance between computational efficiency and detection accuracy, making it suitable for real-time disaster response applications [53].

2) Training Environment

The models were trained with a batch size of 16 over 300 epochs. The input image resolution was fixed at 640 × 640. For reproducibility, all random seeds were set to 0 throughout the experiments [55].

3) Evaluation Metrics

We report mean Average Precision at IoU threshold 0.5 (mAP@0.5) and averaged over multiple thresholds (mAP@0.5:0.95) [56], along with per-class precision and recall, emphasizing `Road-Blocked` due to its operational significance.

Fig. 4 Comprehensive qualitative comparison of generative augmentation methods for `Road-Blocked` synthesis. Each column displays an original `Road-Clear` scene (top row) and its transformation by different methods: InstructPix2Pix text-guided editing (second row), Diffusion with LoRA fine-tuning (third row), successful SGGA dual-guidance generation (fourth row), and typical SGGA failure cases (bottom row). Red boxes highlight generation failures where methods fail to produce convincing road blockages or introduce visual artifacts. Green boxes showcase successful context-aware debris generation that maintains spatial coherence and semantic consistency. SGGA demonstrates superior spatial control and realistic obstruction synthesis compared to text-only (InstructPix2Pix) or global fine-tuning (LoRA) approaches, though occasional failure cases show limitations in complex structural scenarios

4.4.2 Results and Discussion

1) Road-Blocked Class Performance

Experimental results reveal significant differences across augmentation strategies for the underrepresented `Road-Blocked` class. As summarized in Table 4, our SGGA method achieved the highest performance across evaluated metrics. For clarity, all CLIP, LPIPS, and FID values reported in Tables 2 and 3 were computed exclusively from the equally filtered set of 200 samples for each method (SGGA, Diffusion+LoRA, and InstructPix2Pix), ensuring strict experimental fairness.

Relative to the baseline, SGGA achieved a 26.2% improvement in mAP@0.5 (0.336 → 0.424) and a 29.7% gain in recall (0.290 → 0.376), demonstrating effectiveness for rare-class detection.

2) Overall Model Performance

To verify that class-targeted augmentation did not degrade overall performance, we evaluated metrics across all classes (Table 5). Overall mAP remained stable or improved slightly across all methods.

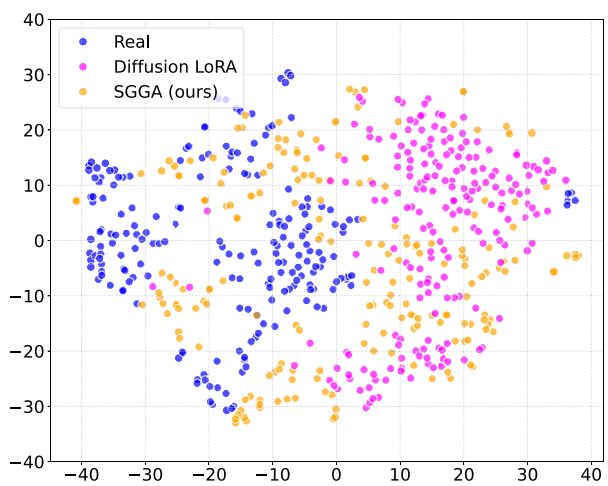

Fig. 5 t-SNE visualization of CLIP embeddings for real and generated `Road-Blocked` images. Blue dots represent real images, magenta dots denote diffusion LoRA-generated images, and orange dots indicate SGGA-generated images

Table 4 Class-specific detection performance for the `Road-Blocked` class

Method	Pr (↑)	Re (↑)	mAP (↑) @0.5/0.5:0.95
No augmentation	0.448	0.290	0.336 / 0.194
Conventional Aug. [51]	0.491	0.344	0.351/0.197
InstructPix2Pix [34]	0.521	0.348	0.389/0.210
Diffusion with LoRA [52]	0.573	0.333	0.373/0.223
SGGA (ours)	0.568	0.376	0.424 / 0.240

Pr Precision, *Re* Recall, *mAP* mAP@0.5/mAP@0.5:0.95

Table 5 Overall detection performance across all classes on RescueNet

Method	mAP (↑) @0.5	mAP (↑) @0.5:0.95
No augmentation	0.725	0.532
Conventional Aug. [51]	0.719	0.518
InstructPix2Pix [34]	0.728	0.525
Diffusion with LoRA [52]	0.733	0.534
SGGA (ours)	0.737	0.532

These results demonstrate that minority-class focused augmentation like SGGA significantly enhances a critical class without harming overall accuracy.

3) SGGA Comparative Analysis and Discussion

SGGA achieved a substantial 26.2% improvement in mAP@0.5 and a 29.7% gain in recall for `Road-Blocked`, underscoring utility for emergency response. Compared to diffusion fine-tuning (LoRA), SGGA adds precise spatial control via semantic masks; compared to instruction-based text editing (InstructPix2Pix), SGGA provides stronger localization and context preservation through dual guidance (mask + prompt). This tailored design explains its strong performance and supports practical deployment.

4.5 Cross-Domain Validation on FloodNet

We mirror the RescueNet pipeline on the FloodNet dataset, targeting the *Road Flooded* class (Road Non-Flooded→Road Flooded) with identical augmentation and training settings. Evaluation follows the same metrics (mAP@0.5 / mAP@0.5:0.95, precision, recall) (Fig. 6).

These results indicate that SGGA's dual guidance generalizes effectively from road blockage to flood scenarios, improving the rare *Road Flooded* class while maintaining overall performance across disaster domains (Tables 6 and 7).

4.6 Ablation Study on the Number of Synthetic Samples

Since determining the optimal number of synthetic samples is inherently difficult, we empirically examined the effect of augmentation scale by adding 0, 50, 100, and 200 SGGA-generated images were added to classes with fewer than 800 original samples. As shown in Table 8, performance consistently improved as the number of synthetic samples increased, but the rate of improvement began to taper off around 200 samples. Although larger-scale augmentation could potentially offer further gains, 200 samples were considered a reasonable and efficient choice given the trade-off between performance improvement and generation cost.

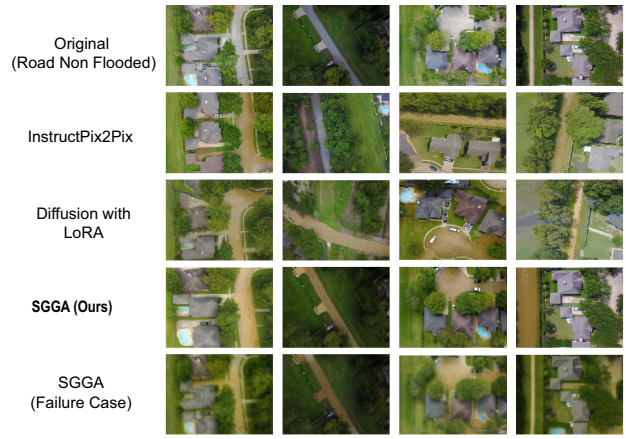

Fig. 6 Cross-domain validation: qualitative comparison of FloodNet post-flood scene understanding. Each column shows an original *Road Non-Flooded* scene (top row) and its transformation by: InstructPix2Pix (second row), Diffusion with LoRA (third row), and SGGA (bottom row). SGGA demonstrates consistent cross-domain performance, generating realistic flood water coverage while preserving road infrastructure and spatial context

Table 6 Road flood detection on FloodNet for the *Road Flooded* class

Method	Pr (↑)	Re (↑)	mAP (↑)@0.5/0.5:0.95
No augmentation	0.463	0.312	0.351/0.204
Conventional Aug	0.487	0.328	0.368/0.217
InstructPix2Pix [34]	0.502	0.345	0.381/0.228
Diffusion with LoRA [52]	0.521	0.352	0.395/0.234
SGGA (ours)	0.567	0.419	0.438/0.261

Pr Precision, *Re* Recall, *mAP* mAP@0.5/mAP@0.5:0.95

Table 7 Overall detection performance across all classes on FloodNet

Method	mAP (↑)@0.5	mAP (↑)@0.5:0.95
No augmentation	0.652	0.441
Conventional Aug	0.659	0.448
InstructPix2Pix [34]	0.665	0.453
Diffusion with LoRA [52]	0.673	0.461
SGGA (ours)	0.684	0.468

5 Conclusion

In this paper, we introduced SGGA (Semantic-Guided Generative Augmentation), a novel framework designed to mit-

Table 8 Ablation study on the number of SGGA-generated samples for `Road-Blocked`

Added SGGA samples	mAP (↑) @0.5	mAP (↑) @0.5:0.95
0	0.336	0.194
50	0.385	0.219
100	0.409	0.231
200	0.424	0.240

igate the critical issue of class imbalance in disaster object detection. Our work presented three main contributions, each validated by our experimental findings.

First, we proposed a novel semantic-guided diffusion framework that leverages segmentation masks for precise, context-aware synthesis. The effectiveness of this framework was confirmed by our image quality assessments, where SGGA-generated samples achieved significantly higher CLIP similarity and lower LPIPS/FID scores compared to semantic-guided baselines including Instruct-Pix2Pix, demonstrating superior semantic fidelity and visual realism.

Second, we integrated SGGA into an end-to-end pipeline with a YOLOv8 detector. Finally, we demonstrated the practical impact of this pipeline through extensive experiments on the highly imbalanced RescueNet dataset and cross-domain validation on FloodNet. Our key finding—a +26.2% improvement in mAP@0.5 and a +29.7% gain in recall for the rare `Road-Blocked` class—directly validates that our approach effectively addresses severe class imbalance without degrading overall model performance. Cross-domain results on FloodNet further confirmed SGGA's generalizability across disaster scenarios.

Future work will first aim to validate the generalizability of SGGA across other critical rare classes in disaster scenarios. For instance, our framework can be directly applied to augment the `Building_Total_Destruction` class by using masks from undamaged buildings as spatial guides and applying damage-specific textual prompts (e.g., "a collapsed building with exposed rebar"). Validating this and other extensions is a primary direction for our future research. Furthermore, we plan to minimize dependency on pixel-level annotations by investigating automated mask generation techniques and extend SGGA beyond UAV imagery to include first-person view (FPV) data from wearable cameras. This expansion will facilitate SGGA's application in diverse disaster contexts, such as building collapses, subway incidents, and forest fire responses.

The final detector, trained with our SGGA-augmented data, is lightweight and thus suitable for real-time deployment on edge devices, supporting rapid situational awareness. Our offline SGGA framework is modular and flexible, making it a practical approach for preparing robust models for a wide range of disaster scenarios in advance.

Appendix A: Visual Evidence of Novel Synthesis

To provide visual evidence that SGGA performs novel synthesis rather than simple feature recombination, Fig. 7 presents a side-by-side comparison. We selected representative patches generated by SGGA and compared them with the most visually similar patches found across the entire original training datasets. The comparison highlights that the textures and shapes synthesized by SGGA are unique and not direct replicas of existing training data, confirming that our method performs creative synthesis.

Fig. 7 Visual comparison of SGGA-generated patches against the most similar patches from the original training sets. The generated patches exhibit novel textures and structures not present in the source data, confirming that SGGA performs creative synthesis rather than mere feature recombination

Ethical Considerations

The deployment of synthetic data generation in disaster response scenarios raises several critical ethical considerations that must be carefully addressed to ensure responsible and safe application.

1) Bias Amplification and Representation

SGGA's reliance on existing disaster datasets introduces the risk of perpetuating and amplifying biases present in the training data. If source datasets under-represent certain geographic regions, disaster types, or demographic contexts, synthetic augmentation may inadvertently reinforce these gaps rather than addressing them. For instance, if training data predominantly contains imagery from developed regions, SGGA-generated samples may not adequately represent disaster scenarios in developing areas with different infrastructure patterns, potentially leading to biased model performance during real-world deployment.

2) Misuse and Misinformation Risks

The high-quality synthetic disaster imagery generated by SGGA could potentially be misused for creating false disaster reports, insurance fraud, or spreading misinformation about disaster events. The realistic nature of SGGA-generated images makes them particularly susceptible to misuse in contexts where visual authenticity is crucial for decision-making. Clear provenance tracking and digital watermarking of synthetic images become essential safeguards against such misuse.

3) Data Authenticity and Transparency

The integration of synthetic data into training pipelines raises questions about data authenticity and the need for transparent documentation. Emergency response systems trained on augmented datasets must clearly distinguish between real and synthetic training data to maintain trust and enable proper validation. Failure to maintain this distinction could compromise the credibility of disaster response systems and potentially impact public trust in automated damage assessment tools.

4) Mitigation Strategies and Responsible Deployment

To address these ethical concerns, we implement several mitigation strategies:

- *Quality Filtering and Human Oversight:* Our multi-stage filtering pipeline (CLIP semantic consistency + LPIPS perceptual similarity) serves as a first line of defense against problematic synthetic content. All generated images undergo manual verification using annotation platforms for critical applications.
- *Transparent Documentation:* Clear labeling and metadata tracking distinguish synthetic from real disaster imagery throughout the training pipeline, enabling proper audit trails and validation procedures.
- *Diverse Source Data:* Future implementations should prioritize diverse and representative training datasets to minimize bias amplification, potentially incorporating imagery from multiple geographic regions and disaster types.
- *Human-in-the-Loop Validation:* Critical disaster response applications should incorporate human oversight mechanisms, particularly for high-stakes decisions involving resource allocation or evacuation planning.

5) Future Ethical Research Directions

Future research should explore adversarial robustness testing to ensure synthetic augmentation does not introduce vulnerabilities, develop automated bias detection mechanisms for synthetic datasets, and establish industry standards for ethical synthetic data generation in safety-critical applications. Additionally, ongoing collaboration with disaster response professionals and affected communities is essential to ensure that synthetic augmentation serves genuine humanitarian needs while respecting the dignity and experiences of disaster-affected populations.

These ethical considerations are fundamental to ensuring that SGGA and similar synthetic augmentation technologies contribute positively to disaster response capabilities while maintaining the highest standards of responsible AI deployment.

Author Contributions DJ created the main idea and led the experiment and technical writing. DH contributed to writing the draft, visualization, ideation, literature review, labeled datasets, and set up the experimental environment. SA conducted the preliminary literature review, labeled additional datasets, and set up and operated the experimental environment. JC provided critical insights regarding disaster scenarios and supervised the research of DJ and SA. SC supervised the overall experiment and technical writing, contributed ideas to generalize the SGGA approach, and assisted with visualization. All authors have read and approved the final manuscript.

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Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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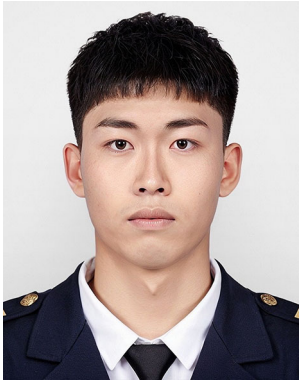

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